Tata Kelola dan Manajemen Teknologi Informasi

Manajemen TI Domain BAI

Team Teaching ISM

S1 Sistem Informasi – Fakultas Rekayasa Industri

Outline

1. BAI01 - Managed **Programs**
2. BAI02 - Managed **Requirements** Definition
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Process and **Organizational Structures** **Components** of BAI Domain

BAI01 - Managed **Programs**

Purpose: **Realize** desired business value and **reduce** the risk of unexpected delays, costs and value erosion.

Practice	Short Description
BAI01.01 Maintain a **standard approach** for program management	Maintain a standard approach for program management that enables governance and management review, decision-making and delivery-management activities.
BAI01.02 **Initiate** a program	Initiate a program to confirm expected benefits and obtain authorization to proceed.
BAI01.03 Manage stakeholder **engagement**	Manage stakeholder engagement to ensure an active exchange of accurate, consistent and timely information for all relevant stakeholders.
BAI01.04 Develop and maintain the **program plan**	Maintain and update the program plan and business case throughout the full economic life cycle of the program, ensuring alignment with strategic objectives and reflecting the current status and insights gained to date.
BAI01.05 **Launch and execute** the program	Launch and execute the program to acquire and direct the resources needed to accomplish the goals and benefits of the program as defined in the program plan.
BAI01.06 **Monitor, control and report** on the program outcomes	Monitor and control performance against plan throughout the full economic life cycle of the investment, covering solution delivery at the program level and value/outcome at the enterprise level.
BAI01.07 Manage program **quality**	Prepare and execute a quality management plan, processes and practices that align with quality management standards (QMS).
BAI01.08 Manage program **risk**	Eliminate or minimize specific risk associated with programs.
BAI01.09 **Close** a program	Remove the program from the active investment portfolio when there is agreement that the desired value has been achieved or when it is clear it will not be achieved within the value criteria set for the program.

BAI01 - Managed **Programs**

Purpose: **Realize** desired business value and **reduce** the risk of unexpected delays, costs and value erosion.

B. Component: Organizational Structures											
Key Management Practice	**Chief Executive Officer**	**Chief Risk Officer**	**Chief Information Officer**	**I&T Governance Board**	**Business Process Owners**	**Steering (Programs/Projects) Committee**	**Program Manager**	**Project Management Office**	**Head Architect**	**Head Development**	**Head IT Operations**
BAI01.01 Maintain a standard approach for program management.	A		R	R			R				
BAI01.02 Initiate a program.		R			R	A	R	R			
BAI01.03 Manage stakeholder engagement.					R	A	R	R			
BAI01.04 Develop and maintain the program plan.						A	R	R			
BAI01.05 Launch and execute the program.			R		R	A	R	R			
BAI01.06 Monitor, control and report on the program outcomes.			R			A	R	R	R	R	R
BAI01.07 Manage program quality.					R	A	R	R			
BAI01.08 Manage program risk.		R			R	A	R	R		R	
BAI01.09 Close a program.			R		R	A	R	R		R	

BAI02 - Managed **Requirements** Definition

Purpose: **Create** optimal solutions that **meet** enterprise needs while minimizing risk.

Practice	Short Description
BAI02.01 Define and maintain business **functional** and **technical requirements**	Based on the business case, identify, prioritize, specify and agree on business information, functional, technical and control requirements.
BAI02.02 Perform a **feasibility study** and **formulate alternative solutions**	Perform a feasibility study of potential alternative solutions, assess their viability and select the preferred option.
BAI02.03 Manage **requirements risk**	Identify, document, prioritize and mitigate functional, technical and information processing-related risk associated with the enterprise requirements, assumptions and proposed solution.
BAI02.04 Obtain **approval** of requirements and solutions	Coordinate feedback from affected stakeholders.

BAI02 - Managed **Requirements** Definition

Purpose: **Create** optimal solutions that **meet** enterprise needs while minimizing risk.

B. Component: Organizational Structures														
Key Management Practice	**Chief Risk Officer**	**Chief Information Officer**	**Business Process Owners**	**Steering (Programs/Projects) Committee**	**Program Manager**	**Project Manager**	**Project Management Office**	**Relationship Manager**	**Head Architect**	**Head Development**	**Head IT Operations**	**Information Security Manager**	**Privacy Officer**
BAI02.01 Define and maintain business functional and technical requirements.			R	A	R	R	R	R	R	R		R	R
BAI02.02 Perform a feasibility study and formulate alternative solutions.			R	A	R	R	R			R			
BAI02.03 Manage requirements risk.	R	R	R	A	R	R	R			R	R	R	R
BAI02.04 Obtain approval of requirements and solutions.			R	A	R	R	R					R	R

BAI03 - Managed **Solutions** Identification and Build

Purpose: Ensure agile and scalable **delivery** of digital **products** and **services**.

Practice	Short Description
BAI03.01 Design **high-level** solutions	Develop and document high-level designs for the solution.
BAI03.02 Design **detailed** solution components	Develop, document and elaborate detailed designs progressively.
BAI03.03 **Develop** solution **components**	Develop solution components progressively in a separate environment, in accordance with detailed designs.
BAI03.04 **Procure** solution **components**	Procure solution components, based on the acquisition plan.
BAI03.05 **Build** solutions	Install and configure solutions and integrate with business process activities.
BAI03.06 Perform **quality assurance** (QA)	Develop, resource and execute a QA plan aligned with the QMS.
BAI03.07 **Prepare** for solution **testing**	Establish a test plan and required environments to test the individual and integrated solution components.
BAI03.08 **Execute** solution **testing**	During development, execute testing continually, in accordance with the defined test plan and development practices in the appropriate environment.
BAI03.09 Manage **changes** to **requirements**	Track the status of individual requirements throughout the project life cycle.
BAI03.10 **Maintain** solutions	Develop and execute a plan for the maintenance of solution and infrastructure components.

BAI03 - Managed **Solutions** Identification and Build

Purpose: Ensure agile and scalable **delivery** of digital **products** and **services**.

Practice	Short Description
BAI03.11 **Define** IT **products and services** and **maintain** the service **portfolio**	Define and agree on new or changed IT products or services and service level options.
BAI03.12 **Design** solutions **based on** the defined development **methodology**	Design, develop and implement solutions with the appropriate development methodology.

BAI03 - Managed **Solutions** Identification and Build

Purpose: Ensure agile and scalable **delivery** of digital **products** and **services**.

B. Component: Organizational Structures																			
Key Management Practice	**Chief Information Officer**	**Chief Technology Officer**	**Chief Digital Officer**	**Business Process Owners**	**Portfolio Manager**	**Steering (Programs/Projects) Committee**	**Program Manager**	**Project Manager**	**Project Management Office**	**Relationship Manager**	**Head Architect**	**Head Development**	**Head IT Operations**	**Head IT Administration**	**Service Manager**	**Information Security Manager**	**Business Continuity Manager**	**Privacy Officer**
BAI03.01 Design high-level solutions.		R		R		A	R	R	R	R		R				R		
BAI03.02 Design detailed solution components.		R		R		A	R	R	R			R						
BAI03.03 Develop solution components.		R		R		A	R	R	R			R						
BAI03.04 Procure solution components.		R		R		A						R	R	R				
BAI03.05 Build solutions.		R		R		A	R	R	R			R				R		
BAI03.06 Perform quality assurance (QA).		R		R		A	R	R	R			R						
BAI03.07 Prepare for solution testing.		R		R		A						R	R		R	R	R	R
BAI03.08 Execute solution testing.		R		R		A						R	R			R		R
BAI03.09 Manage changes to requirements.		R		R		A	R	R	R		R	R				R		R
BAI03.10 Maintain solutions.	A	R		R			R	R	R			R				R		R
BAI03.11 Define IT products and services and maintain the service portfolio.	A														R	R		R
BAI03.12 Design solutions based on the defined development methodology.	A		R		R		R	R										

BAI04 - Managed **Availability** and **Capacity**

Purpose: **Maintain** service availability, efficient management of **resources** and optimization of **system performance** through **prediction** of future performance and capacity requirements.

Practice	Short Description
BAI04.01 Assess **current** availability, performance and capacity and create a **baseline**	Assess availability, performance and capacity of services and resources to ensure that cost-justifiable capacity and performance are available to support business needs and deliver against service level agreements (SLAs).
BAI04.02 Assess **business impact**	Identify important services to the enterprise.
BAI04.03 **Plan** for **new or changed** service **requirements**	Plan and prioritize availability, performance and capacity implications of changing business needs and service requirements.
BAI04.04 **Monitor and review** availability and capacity	Monitor, measure, analyze, report and review availability, performance and capacity.
BAI04.05 **Investigate and address** availability, performance and capacity **issues**	Address deviations by investigating and resolving identified availability, performance and capacity issues.

BAI04 - Managed **Availability** and **Capacity**

Purpose: **Maintain** service availability, efficient management of **resources** and optimization of **system performance** through **prediction** of future performance and capacity requirements.

B. Component: Organizational Structures								
Key Management Practice	**Executive Committee**	**Chief Information Officer**	**Chief Technology Officer**	**Business Process Owners**	**Head Architect**	**Head IT Operations**	**Service Manager**	**Business Continuity Manager**
BAI04.01 Assess current availability, performance and capacity and create a baseline.		R	A	R		R	R	
BAI04.02 Assess business impact.	A			R		R	R	
BAI04.03 Plan for new or changed service requirements.		R	A	R		R	R	
BAI04.04 Monitor and review availability and capacity.	A			R		R	R	
BAI04.05 Investigate and address availability, performance and capacity issues.		R	A	R	R	R	R	R

BAI05 - Managed **Organizational Change**

Purpose: **Prepare** and **commit** stakeholders for business change and **reduce** the risk of failure.

Practice	Short Description
BAI05.01 Establish the **desire** to change	Understand the scope and impact of the desired change.
BAI05.02 Form an effective **implementation team**	Establish an effective implementation team by assembling appropriate members, creating trust, and establishing common goals and effectiveness measures.
BAI05.03 **Communicate** desired vision	Communicate the desired vision for the change in the language of those affected by it.
BAI05.04 **Empower role players** and identify short-term wins	Empower those with implementation roles by assigning accountability.
BAI05.05 Enable **operation and use**	Plan and implement all technical, operational and usage aspects so all those who are involved in the future state environment can exercise their responsibility.
BAI05.06 Embed **new approaches**	Embed new approaches by tracking implemented changes, assessing the effectiveness of the operation and use plan, and sustaining ongoing awareness through regular communication.
BAI05.07 **Sustain changes**	Sustain changes through effective training of new staff, ongoing communication campaigns, continued commitment of top management, monitoring of adoption and sharing of lessons learned across the enterprise.

BAI05 - Managed **Organizational Change**

Purpose: **Prepare** and **commit** stakeholders for business change and **reduce** the risk of failure.

B. Component: Organizational Structures																		
Key Management Practice	Executive Committee	Chief Executive Officer	Chief Operating Officer	Chief Information Officer	Chief Technology Officer	Chief Digital Officer	I&T Governance Board	Business Process Owners	Program Manager	Project Manager	Project Management Office	Head Human Resources	Head Development	Head IT Operations	Service Manager	Information Security Manager	Business Continuity Manager
BAI05.01 Establish the desire to change.	R	A		R	R	R	R	R	R	R		R					
BAI05.02 Form an effective implementation team.	A			R	R	R			R	R	R		R				
BAI05.03 Communicate desired vision.	A			R	R	R	R		R	R							
BAI05.04 Empower role players and identify short-term wins.	A			R	R	R			R	R		R					
BAI05.05 Enable operation and use.	A		R	R	R	R		R			R		R	R	R	R	R
BAI05.06 Embed new approaches.	A		R	R	R	R		R			R		R	R	R	R	R
BAI05.07 Sustain changes.	A		R	R	R	R		R	R	R	R		R	R	R	R	R

BAI06 - Managed **IT Changes**

Purpose: Enable fast and reliable **delivery of change** to the business.

Practice	Short Description
BAI06.01 Evaluate, prioritize and authorize **change requests**	Evaluate all requests for change to determine the impact on business processes and I&T services, and to assess whether change will adversely affect the operational environment and introduce unacceptable risk.
BAI06.02 Manage **emergency changes**	Carefully manage emergency changes to minimize further incidents.
BAI06.03 Track and report **change status**	Maintain a tracking and reporting system to document rejected changes and communicate the status of approved, in-process and complete changes.
BAI06.04 **Close and document** the changes.	Whenever changes are implemented, update the solution, user documentation and procedures affected by the change.

BAI06 - Managed **IT Changes**

Purpose: Enable fast and reliable **delivery of change** to the business.

B. Component: Organizational Structures										
Key Management Practice	**Chief Information Officer**	**Business Process Owners**	**Program Manager**	**Project Manager**	**Head Development**	**Head IT Operations**	**Service Manager**	**Information Security Manager**	**Business Continuity Manager**	**Privacy Officer**
BAI06.01 Evaluate, prioritize and authorize change requests.	A	R			R	R	R	R	R	R
BAI06.02 Manage emergency changes.	A				R	R	R	R		R
BAI06.03 Track and report change status.	A	R	R	R	R	R	R			
BAI06.04 Close and document the changes.	A	R	R	R	R	R	R		R	

BAI07 - Managed IT Change Acceptance and **Transitioning**

Purpose: **Implement solutions** safely and **in line** with the agreed expectations and outcomes.

Practice	Short Description
BAI07.01 Establish an **implementation plan**	Establish an implementation plan that covers system and data conversion, acceptance testing criteria, communication, training, release preparation, promotion to production, early production support, a fallback/back-up plan, and a post-implementation review.
BAI07.02 **Plan** business process, system and data **conversion**	Prepare for business process, I&T service data and infrastructure migration as part of the enterprise's development methods.
BAI07.03 **Plan** acceptance **tests**	Establish a test plan based on enterprisewide standards that define roles, responsibilities, and entry and exit criteria.
BAI07.04 Establish a **test environment**	Define and establish a secure test environment representative of the planned business process and IT operations environment.
BAI07.05 Perform **acceptance tests**	Test changes independently, in accordance with the defined test plan, prior to migration to the live operational environment.
BAI07.06 **Promote to production** and manage releases	Promote the accepted solution to the business and operations.
BAI07.07 Provide **early production support**	For an agreed period of time, provide early support to users and I&T operations to resolve issues and help stabilize the new solution.
BAI07.08 Perform a **post-implementation review**	Conduct a post-implementation review to confirm outcome and results, identify lessons learned, and develop an action plan.

BAI07 - Managed IT Change Acceptance and **Transitioning**

Purpose: **Implement solutions** safely and **in line** with the agreed expectations and outcomes.

B. Component: Organizational Structures									
Key Management Practice	**Chief Information Officer**	**Business Process Owners**	**Data Management Function**	**Head Development**	**Head IT Operations**	**Service Manager**	**Information Security Manager**	**Business Continuity Manager**	**Privacy Officer**
BAI07.01 Establish an implementation plan.	A	R		R		R	R	R	
BAI07.02 Plan business process, system and data conversion.	A	R	R	R		R	R	R	
BAI07.03 Plan acceptance tests.	A	R		R	R		R	R	R
BAI07.04 Establish a test environment.	A	R		R	R		R	R	
BAI07.05 Perform acceptance tests.	A	R		R	R		R	R	R
BAI07.06 Promote to production and manage releases.	A	R		R	R	R		R	
BAI07.07 Provide early production support.	A	R		R	R	R			
BAI07.08 Perform a post-implementation review.	A	R		R	R	R			

BAI08 - Managed **Knowledge**

Purpose: **Provide** the **knowledge and information** required **to support** all staff in the governance and management of enterprise I&T and **allow** for informed decision making.

Practice	Short Description
BAI08.01 **Identify and classify sources of information** for governance and management of I&T	Identify, validate and classify diverse sources of internal and external information required to enable governance and management of I&T.
BAI08.02 **Organize** and **contextualize information** into knowledge	Organize information based on classification criteria.
BAI08.03 **Use and share** knowledge	Propagate available knowledge resources to relevant stakeholders and communicate how these resources can be used to address different needs.
BAI08.04 **Evaluate and update or retire** information	Measure the use and evaluate the currency and relevance of information. Update information or retire obsolete information.

BAI08 - Managed **Knowledge**

Purpose: **Provide** the **knowledge and information** required **to support** all staff in the governance and management of enterprise I&T and **allow** for informed decision making.

B. Component: Organizational Structures																			
Key Management Practice	**Chief Information Officer**	**Chief Technology Officer**	**Chief Digital Officer**	**Business Process Owners**	**Portfolio Manager**	**Program Manager**	**Project Manager**	**Data Management Function**	**Head Architect**	**Head Development**	**Head IT Operations**	**Head IT Administration**	**Service Manager**	**Information Security Manager**	**Business Continuity Manager**	**Privacy Officer**	**Legal Counsel**
BAI08.01 Identify and classify sources of information for governance and management of I&T.	A			R				R		R	R		R				
BAI08.02 Organize and contextualize information into knowledge.	A							R		R	R	R					
BAI08.03 Use and share knowledge.	A	R	R	R	R	R	R	R				R					R
BAI08.04 Evaluate and update or retire information.	A			R		R	R	R	R	R	R	R	R	R	R	R	

BAI09 - Managed **Assets**

Purpose: **Account** for all I&T assets and **optimize** the value provided by their use.

Practice	Short Description
BAI09.01 **Identify and record** current assets	Maintain an up-to-date, accurate record of all I&T assets that are required to deliver services and that are owned or controlled by the organization
BAI09.02 Manage **critical assets**	Identify assets that are critical in providing service capability. Maximize their reliability and availability to support business needs.
BAI09.03 Manage the asset **life cycle**	Manage assets from procurement to disposal.
BAI09.04 Optimize asset **value**	Regularly review the overall asset base to identify ways to optimize value in alignment with business needs.
BAI09.05 Manage **licenses**	Manage software licenses to maintain the optimal number of licenses and support business requirements.

BAI09 - Managed **Assets**

Purpose: **Account** for all I&T assets and **optimize** the value provided by their use.

B. Component: Organizational Structures									
Key Management Practice	**Chief Information Officer**	**Chief Technology Officer**	**Head Architect**	**Head Development**	**Head IT Operations**	**Head IT Administration**	**Service Manager**	**Information Security Manager**	**Privacy Officer**
BAI09.01 Identify and record current assets.		A			R	R			
BAI09.02 Manage critical assets.		A	R	R	R	R		R	R
BAI09.03 Manage the asset life cycle.		A			R	R	R		
BAI09.04 Optimize asset value.	A	R	R	R	R	R	R		
BAI09.05 Manage licenses.	A	R		R	R	R			

BAI10 - Managed **Configuration**

Purpose: **Provide** sufficient information about service assets to **enable the service** to be effectively managed.

Practice	Short Description
BAI10.01 Establish and maintain a **configuration model**	Establish and maintain a logical model of the services, assets, infrastructure and recording of configuration items (CIs), including the relationships among them.
BAI10.02 Establish and maintain a **configuration repository** and baseline	Establish and maintain a configuration management repository and create controlled configuration baselines.
BAI10.03 Maintain and control **configuration items**	Maintain an up-to-date repository of configuration items (CIs) by populating any configuration changes.
BAI10.04 Produce status and **configuration reports**	Define and produce configuration reports on status changes of configuration items.
BAI10.05 **Verify and review** integrity of the **configuration repository**	Periodically review the configuration repository and verify completeness and correctness against the desired target.

BAI10 - Managed **Configuration**

Purpose: **Provide** sufficient information about service assets to **enable the service** to be effectively managed.

B. Component: Organizational Structures								
Key Management Practice	**Chief Information Officer**	**Chief Technology Officer**	**Head Architect**	**Head Development**	**Head IT Operations**	**Head IT Administration**	**Service Manager**	**Information Security Manager**
BAI10.01 Establish and maintain a configuration model.		A			R	R	R	
BAI10.02 Establish and maintain a configuration repository and baseline.		A		R	R	R	R	R
BAI10.03 Maintain and control configuration items.	A	R		R	R	R		
BAI10.04 Produce status and configuration reports.		A			R	R		
BAI10.05 Verify and review integrity of the configuration repository.		A	R	R	R		R	

BAI11 - Managed **Projects**

Purpose: **Realize** defined project outcomes and **reduce** the risk of unexpected delays, costs and value erosion.

Practice	Short Description
BAI11.01 Maintain a **standard approach** for project management	Maintain a standard approach for project management that enables governance and management review, decision-making and delivery-management activities.
BAI11.02 **Start up and initiate** a project	Define and document the nature and scope of the project to confirm and develop a common understanding of project scope among stakeholders.
BAI11.03 Manage stakeholder **engagement**	Manage stakeholder engagement to ensure an active exchange of accurate, consistent and timely information that reaches all relevant stakeholders.
BAI11.04 Develop and maintain the **project plan**	Establish and maintain a formal, approved, integrated project plan (covering business and IT resources) to guide project execution and control throughout the life of the project.
BAI11.05 Manage project **quality**	Prepare and execute a quality management plan, processes and practices that align with quality management standards (QMS).
BAI11.06 Manage project **risk**	Eliminate or minimize specific risk associated with projects.
BAI11.07 **Monitor and control** projects	Measure project performance against key project performance criteria such as schedule, quality, cost and risk.
BAI11.08 Manage project **resources** and **work packages**	Manage project work packages by placing formal requirements on authorizing and accepting work packages and assigning and coordinating appropriate business and IT resources.
BAI11.09 **Close** a project or iteration	At the end of each project, release or iteration, require the project stakeholders to ascertain whether the project, release or iteration delivered the required results in terms of capabilities and contributed as expected to program benefits.

BAI11 - Managed **Projects**

Purpose: **Realize** defined project outcomes and **reduce** the risk of unexpected delays, costs and value erosion.

B. Component: Organizational Structures											
Key Management Practice	Chief Executive Officer	Chief Risk Officer	Chief Information Officer	Chief Technology Officer	Business Process Owners	Steering (Programs/Projects) Committee	Program Manager	Project Manager	Project Management Office	Head Development	Information Security Manager
BAI11.01 Maintain a standard approach for project management.	A		R				R	R			
BAI11.02 Start up and initiate a project.		R		R	R	A	R	R	R	R	
BAI11.03 Manage stakeholder engagement.			R			A		R			
BAI11.04 Develop and maintain the project plan.						A		R	R		
BAI11.05 Manage project quality.		R	R			A		R			R
BAI11.06 Manage project risk.			R			A		R			R
BAI11.07 Monitor and control projects.					R	A		R	R	R	
BAI11.08 Manage project resources and work packages.					R	A	R		R	R	
BAI11.09 Close a project or iteration.						A		R	R		

Information Input Example of BAI Domain

Information Input Example of BAI Domain

Objective	Practice	Input	Description
BAI01 - Managed **Programs**	BAI01.02 Initiate a program	APO05.02	Program **business case**
BAI02 - Managed **Requirements Definition**	BAI02.01 Define and maintain business functional and technical requirements	APO11.02	Acceptance **criteria**
BAI03 - Managed **Solutions** Identification and Build	BAI03.01 Design high-level solutions	APO04.04	Evaluations of **innovation ideas**
BAI04 - Managed **Availability** and **Capacity**	BAI04.01 Assess current availability, performance and capacity and create a baseline	BAI02.01	**Requirements definition** repository
BAI05 - Managed **Organizational Change**	BAI05.01 Establish the desire to change	APO11.02	Results of **quality of service**, including customer feedback
BAI06 - Managed **IT Changes**	BAI06.01 Evaluate, prioritize and authorize change requests	DSS02.03	Approved **service requests**
BAI07 - Managed IT Change Acceptance and **Transitioning**	BAI07.08 Perform a post-implementation review	APO11.03	Results of solution and service delivery **quality monitoring**
BAI08 - Managed **Knowledge**	BAI08.03 Use and share knowledge	BAI05.07	Knowledge **transfer plans**
BAI09 - Managed **Assets**	BAI09.01 Identify and record current assets	BAI10.02	**Configuration** repository
BAI10 - Managed **Configuration**	BAI10.02 Establish and maintain a configuration repository and baseline	BAI09.05	Register of **software licenses**
BAI11 - Managed **Projects**	BAI11.09 Close a project or iteration	BAI07.08	**Post-implementation review** report

Thank You

That’s all